Project Overview

This project analyzes the e-commerce performance of **Toy Store** using transaction and web session data sourced from **Maven Analytics**. The analysis evaluates core business functions including revenue growth across 2012–2015, product portfolio profitability, marketing channel conversion rates, repeat traffic behaviors, and product refund/return impact.

Business Questions

1. Which products drive the vast majority of total revenue and profit margin?
2. How has revenue and average order revenue evolved month-over-month and year-over-year?
3. Which marketing acquisition channels (UTM sources) deliver the highest volume and conversion rates?
4. Do repeat web session visitors convert at a higher rate compared to new visitors?
5. What is the total refund percentage across orders, and which products generate the highest refund liabilities?

SQL Skills Used

- **Aggregations & Grouping:** SUM(), AVG(), COUNT(), COUNT(DISTINCT), GROUP BY, HAVING

- **Window Functions:** ROW_NUMBER() OVER(PARTITION BY ... ORDER BY ...)

- **Joins:** INNER JOIN, LEFT JOIN
- **Conditional Logic & Data Cleaning:** CASE WHEN, COALESCE(), NULLIF()

- **Common Table Expressions (CTEs):** WITH CTE_Name AS (...)
- **Type Casting & Formatting:** CAST(), ROUND(), DECIMAL

Analysis & Results

1. Overall Revenue Growth & Seasonality

- **Yearly Average Revenue:** Average order revenue increased from **$49.99** in 2012 to **$63.80** in 2014, stabilizing at **$62.80** in early 2015.
- **Peak Revenue Months:** Performance steadily grew from **$2,999.40** in March 2012 to a peak of **$144,823.02** in December 2014. The end-of-year holiday surge (November–December) consistently drives maximum volume.

2. Product Revenue Rank & Margins

- **The Original Mr. Fuzzy:** Dominates the portfolio across all years (2012–2015).
  - Total Volume: **29,618 units sold** across **23,861 orders**

  - Total Revenue: **$1,419,767.87**

    - Net Profit: **$879,952.06** (Avg Selling Price: **$59.50**, Cost: **$22.62**)
- **Secondary Products:**
    - **The Forever Love Bear: $318,109.19** total revenue (**$200,348.01** profit).
    - **The Birthday Sugar Panda: $180,857.04** total revenue (**$122,410.51** profit).
    - **The Hudson River Mini Bear: $19,775.72** total revenue (**$13,429.00** profit).

3. Revenue Classification Category

- **High Revenue (> $500k):** The Original Mr. Fuzzy
- **Medium Revenue ($100k - $500k):** The Forever Love Bear, The Birthday Sugar Panda
- **Low Revenue (< $100k):** The Hudson River Mini Bear

4. Marketing Channel Breakdown & Traffic Source Conversion

- **Traffic Distribution:** gsearch leads paid volume with **316,035 sessions** (6% conversion rate). bsearch generated **62,823 sessions** (7% conversion rate). Unknown channels accounted for **83,328 sessions**.
- **Social Channels:** socialbook drove **10,685 sessions** but yielded the lowest conversion rate at **3%**.

5. Session Type Behavior (New vs. Repeat)

- **New Visitors (is_repeat_session = 0): 6.64%** conversion rate.
- **Repeat Visitors (is_repeat_session = 1): 7.83%** conversion rate.
- **Repeat Visitor Traffic:** Unknown sources account for **62%** of repeat traffic.

6. Product Refund Analysis

- **Overall Refund Rate: 5%** of all unique orders incurred a refund.
- **Refund Values:**
    - **The Original Mr. Fuzzy:** Highest absolute dollar refund (**$61,837.63**) due to sales volume.
    - **The Birthday Sugar Panda: $13,842.99** refunded.
    - **The Forever Love Bear: $7,738.71** refunded.
    - **The Hudson River Mini Bear: $1,919.36** refunded.

Key Insights

- **Product Concentration Risk:** "The Original Mr. Fuzzy" accounts for over **70% of total profit**. Expanding marketing focus on mid-tier products like "The Forever Love Bear" can diversify revenue risks.
- **Repeat Visitor Value:** Repeat visitors convert at a higher rate (**7.83% vs 6.64%**). Retaining users and bringing them back generates higher organic value.
- **Marketing Efficiency:** socialbook has a low 3% conversion rate and 0% repeat session contribution. Reallocating budget toward search networks (gsearch / bsearch) will maximize acquisition ROI.